Wing Theatre Controller

User Manual

Show-control software for the Behringer Wing digital mixer, built for theatre and live production.

Covers version 1.9.9 · October 2026

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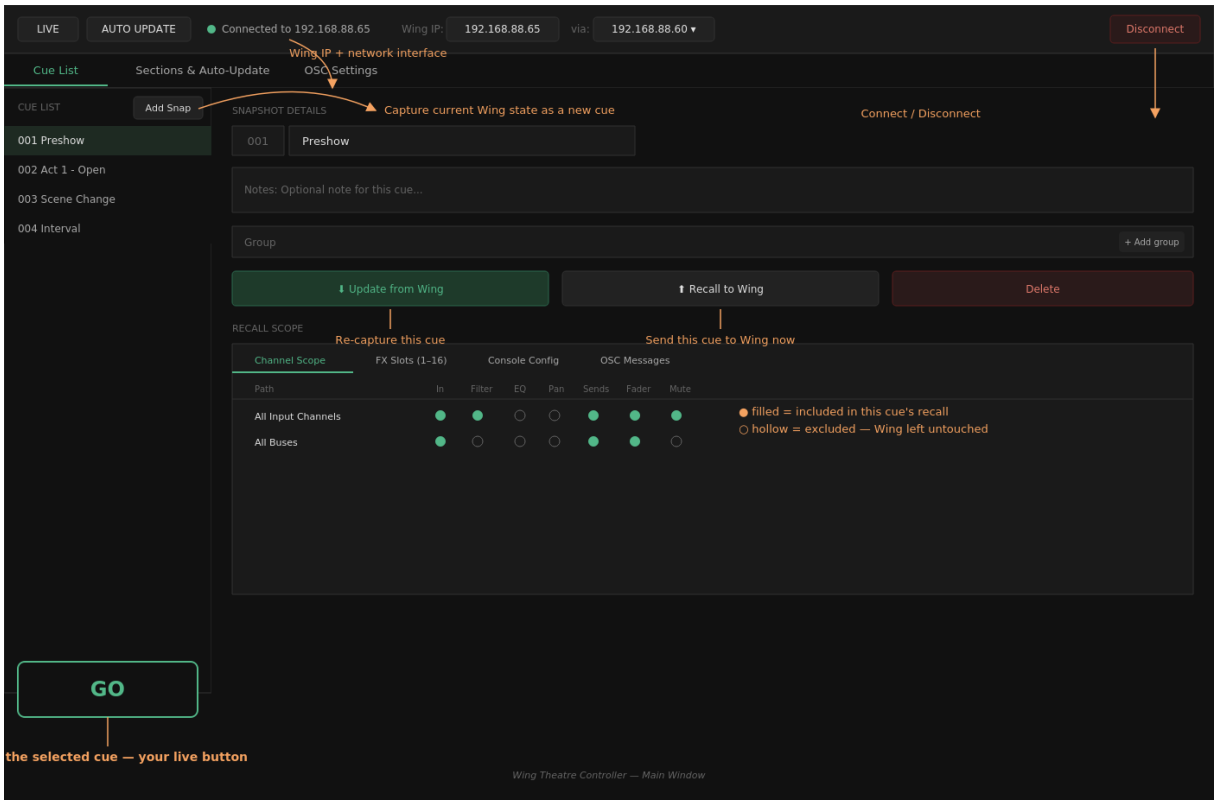
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1. What This Software Does

Wing Theatre Controller connects to a Behringer Wing over the network and lets you:

- Capture snapshots of the console's current state
- Build a cue list and fire snapshots in sequence with a single GO
- Choose exactly which parameters each snapshot recalls (Recall Scope) — down to individual bus and matrix sends — so a cue can change only the vocal mics, only the band, only FX, etc.
- Fade faders and send levels smoothly instead of jumping, with a per-channel fade time and a Cancel fade button
- Keep snapshots automatically up to date while you mix, using Auto Update
- Run the show from a compact LIVE view
- Trigger cues remotely via OSC or from Bitfocus Companion (with two-way status feedback)

It runs alongside the Wing, not instead of it — you still mix on the console or in Wing Editor as normal. Wing Theatre Controller listens to everything you do and remembers it in snapshots.



Main window overview — diagrams in this manual are illustrative layouts matching the app's real colour scheme; exact spacing may differ slightly from your installed version.

2. Installation

macOS

- Download WingTheatreController-macos.dmg from the Releases page.
- Open the DMG and drag Wing Theatre Controller into Applications.

- First launch: the app is not notarised by Apple, so macOS will block the first launch. Right-click the app → Open → Open. On newer macOS versions where that option is missing: try to open the app once, then go to System Settings → Privacy & Security, scroll down and click Open Anyway. This is a one-time step per computer.
- When asked, allow the app to access devices on your local network — without this it cannot reach the Wing.

Everything the app needs (including the wingmon helper that talks to the Wing) is inside the app bundle — there is nothing else to install.

Windows

- Download WingTheatreController-windows.zip and extract it anywhere (e.g. C:\WingTheatre\). Use Windows' own Extract All — the zip is meant for Windows and may not open correctly with the macOS Archive Utility.
- Run Wing Theatre Controller.exe. Keep the whole extracted folder together — the exe needs the files next to it.
- If Windows Firewall prompts you the first time you connect or enable the Companion TCP server, click Allow access.

No separate installation of Wing drivers or software is required — Wing Theatre Controller talks to the console directly over the network.

Install the app in a folder you can write to (Applications, your user folder, C:\WingTheatre\...). The log files described in §13 are written next to the app.

3. Connecting to the Wing

The connection bar sits just below the toolbar:

Field	Purpose
LIVE	Switches to the compact show-running view (see §5)
AUTO UPDATE	Turns Auto Update on/off (see §8)
Status dot	Red = not connected, green = connected; the text next to it shows the state
Wing IP	The Wing console's IP address on your network
via	Which network interface on your computer to use — important if your machine has more than one active network connection (Wi-Fi + Ethernet, VPN, etc.). The small button next to it refreshes the list.
Connect / Disconnect	Opens or closes the connection

Connecting and syncing

Click **Connect**. The app first connects, then reads the Wing's complete current state. The status bar shows **⟳ Syncing Wing state — please wait...** and then **✓ Wing connected — N parameters synced** (typically several thousand parameters, a few seconds).

Actions are locked until the sync is complete

Everything that reads from or writes to the console is disabled until the app is connected **and** the sync has finished:

- GO and Recall to Wing

- Update from Wing
- Add Snap

This also applies when these are triggered remotely (Space bar, OSC, Companion). If you try anyway, the status bar explains why, e.g. **GO blocked — still syncing with the Wing — wait a moment** or **GO blocked — not connected to a Behringer Wing**.

Everything else works offline: you can open a show and edit Recall Scope, fade times, names, notes, groups, sections and OSC settings without a console.

If the sync isn't confirmed within 15 seconds but more than 1000 parameters have arrived, the actions are unlocked anyway with a warning — being locked mid-show would be worse. If too little arrived, they stay locked: disconnect and connect again.

If the connection is lost

If the network to the Wing drops, the status dot turns red, the actions lock again, and the message **wingmon disconnected — Wing connection lost** is shown and logged. Once the network is back, press **Connect** and wait for the sync to finish before pressing GO.

If it won't connect

- Confirm the Wing IP is correct and reachable (ping it)
- Check you selected the right network interface in via
- On macOS, check that the app is allowed Local Network access (System Settings → Privacy & Security → Local Network)
- Look in wing_theatre_log.txt (see §13) for the exact error

4. The Cue List

The left-hand panel is your show's running order.

- **Add Snap** — asks for a name and captures the Wing's current state as a new cue at the end of the list. The new cue immediately becomes the active cue, so Auto Update writes to it straight away. (Only available when connected and synced.)
- Click a cue to select it and see its Snapshot Details on the right
- Drag to reorder cues
- Shift/Cmd/Ctrl-click to select several cues
- Each cue has a number, an editable name, an optional group and optional Notes

Right-click menu

- **Duplicate** — makes a fully independent copy of the cue (one cue selected)
- **Delete cue / Delete N cues** — asks for confirmation (Delete or Backspace does the same)
- **Set group tag... / Clear group tag** — assign one or many cues to a group (see §9)

Snapshot Details panel

- **Update from Wing** — overwrites this cue's stored values with the console's current state. It is instant: the app already knows the complete live state from the sync and every change since, so it doesn't need to ask the Wing again. The cue's Recall Scope and fade times are not changed.

- **Recall to Wing** — sends this cue to the console immediately, respecting its Recall Scope and fades. Unlike GO, it does not make the cue the active cue and does not move the selection.
- **Delete** — removes the cue
- **Group** — pick an existing group or use + Add group (see §9)
- **Notes** — free text; shown in the notes bar in LIVE mode

GO

The big **GO** button (bottom-left) recalls the currently selected cue to the Wing, makes it the active cue, sends its OSC messages and moves the selection to the next cue. This is the button you'll use live. It respects that cue's Recall Scope and fade times.

Cancel fade

Next to GO is **Cancel fade**. While fades are running it shows the time left on the newest fade, e.g. **Cancel fade (12s)**. Pressing it stops all running fades and jumps every fading parameter straight to its target value. See §7.

Keyboard shortcuts

Key	Action
Space	GO
Ctrl+S / Cmd+S	Save
Ctrl+A / Cmd+A	Add Snap
Delete / Backspace	Delete selected cue(s), with confirmation

5. LIVE Mode

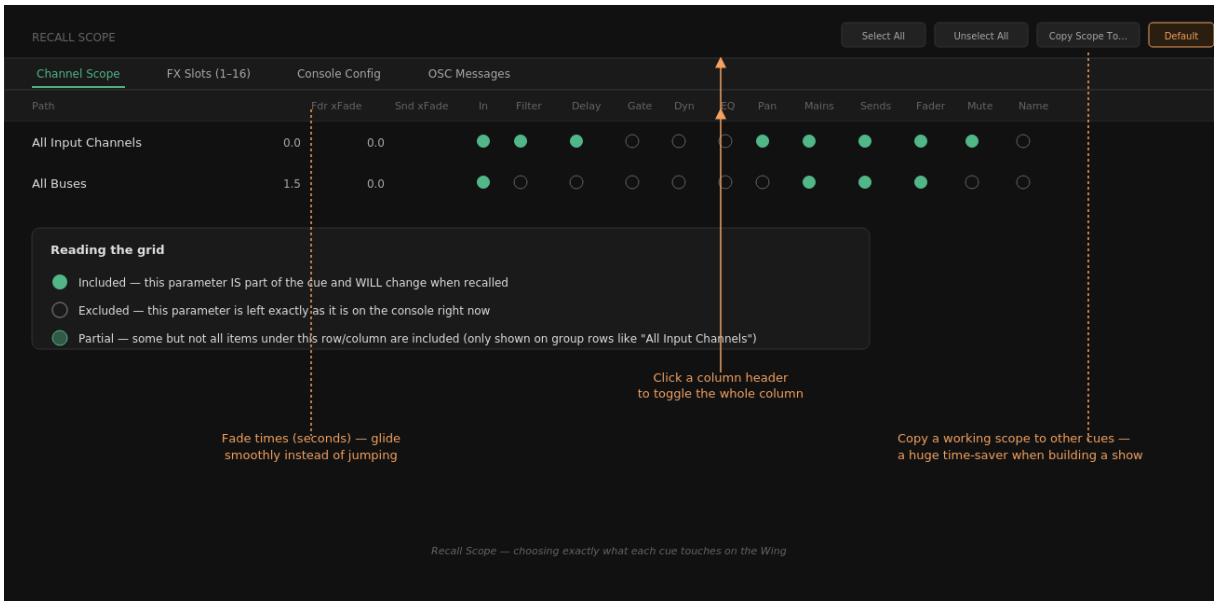
Click **LIVE** in the connection bar to switch to a compact view for running the show:

- The window narrows to a cue list with GO and Cancel fade
- Snapshot Details, the other tabs, the IP fields and Add Snap are hidden, so nothing can be edited by accident
- The notes bar shows the Notes of the selected cue
- The LIVE button turns green and the window title ends in — LIVE

Click LIVE again to return to the full editing view at the window size you had before. Saving while in LIVE mode stores the normal (full) window size, not the narrow live window.

6. Recall Scope — Choosing What Each Cue Changes

This is the heart of the software. Every cue has its own Recall Scope: a per-parameter map of what gets sent to the Wing and what's left untouched.



The Recall Scope grid (illustrative)

Four tabs under Recall Scope:

Tab	Covers
Channel Scope	Input channels, buses, matrix outputs, main outputs and DCA groups — broken down by function: Conn, In, Filter, Delay, Gate, Dyn, Pre Ins, Post Ins, EQ, Pan, Mains, Sends, Fader, Mute, Cfg, Name, Tags
FX Slots (1–16)	The 16 internal FX processors
Console Config	Global console settings
OSC Messages	Custom OSC messages sent to external gear when this cue recalls (with a Send Test button)

Each cell is a toggle:

- ● Included (solid green) — this parameter is part of the cue and will be recalled
- ○ Excluded (hollow) — this parameter is ignored; the console stays as it is
- ◐ Partial — some but not all items in that row/column are included
- — (dash) — not applicable, e.g. EQ on a DCA (DCAs only have Fader, Mute and Name)

Rows and columns

- Click a **column header** (e.g. "EQ") to toggle that column for every row
- Click a circle on a **group row** (e.g. "All Input Channels") to set it for all channels in that group
- Click the ▶ arrow on a group row to expand it to individual channels

Channel names

Individual rows show the console's name for each strip, e.g. **Input Ch 01 (BD)**. If a channel has no name of its own, the name of the input source it is patched to is shown — the same name the Wing shows on the strip. Names update live when you rename on the console. When you're offline, the names stored in the selected cue are used.

Frozen columns

Path, the expand arrow and the two fade columns (Fdr xFade, Snd xFade) stay fixed on the left while you scroll sideways, so you always know which row you're on.

Individual bus and matrix sends

Click the ▶ **Sends** header to fold out one column per destination: Bus 1–16 and Matrix 1–8 (shown as Bus1…Bus16, Mtx1…Mtx8). Click ▼ **Sends** to fold them away again.

- Each circle controls whether that channel's send level/mute to that one bus or matrix is recalled — e.g. recall the monitor sends but leave the FX sends alone
- Click a destination header (e.g. Bus3) to toggle that send for every input and bus
- The main Sends circle shows ◐ when only some sends are included; clicking it sets all sends on or off at once
- Per-send columns apply to input channels and buses (matrices, mains and DCAs have no sends)

Toolbar buttons

- **Select All / Unselect All** — set every cell in the current tab
- **Copy Scope To…** — copy this cue's scope to other cues (huge time-saver for building a show)
- **Default** — left-click resets this cue's scope to a 1-to-1 copy of the app's current default scope

Setting your own default scope

Right-click the **Default** button to open "Edit Default Recall Scope." This is a full scope editor — the same grid as on a normal cue — where you set what the default should be, including per-group differences (e.g. Sends off for buses only) and default fade times. FX Slots and Console Config have their own defaults too; OSC Messages are never part of the default.

Every new cue — including ones created with Add Snap — starts out with whatever the current default scope is. Set it once to match how you like to work. Changes in this dialog only affect cues where you explicitly press Default (and new cues) — existing cues are never changed automatically. The default scope is saved in the show file.

Why this matters

A cue with only Mute and Fader included on "All Input Channels" will only move faders and mutes when recalled — EQ, dynamics, pan, everything else on the console is left completely alone. This lets you build cues that touch just what a scene actually needs.

How a recall is sent

A recall is sent to the Wing in one ordered batch: mutes and faders first, then everything else. Within a section, settings that others depend on go first — e.g. the EQ/Dynamics model is set before that model's own parameters, and send mode before send level — so model-specific values (Dual Dynamic EQ etc.) land correctly.

7. Fades

Two fade-time columns appear per row in Channel Scope:

- **Fdr xFade** — fade time (seconds) for that channel/bus's fader level
- **Snd xFade** — fade time (seconds) for its send levels

Set a fade time greater than 0.0 and that parameter glides to its target over that many seconds when the cue recalls, instead of jumping.

Setting fade times

- Double-click a fade cell and type a value — e.g. 20, 20.0 or 20,5 (comma or period both work). Press Enter or click elsewhere to apply.
- A value on a **group row** (e.g. All Input Channels) is the fade time for every channel in that group
- A value on an **individual channel** overrides the group for that channel only. An empty channel cell means "use the group's time"
- If something can't be read as a number, the status bar says so and the cell resets to 0 (inherit from group)

How fades move

Fades run at 40 steps per second and are sent to the Wing as one combined message per step, no matter how many faders are moving — this keeps network load low even for long fades on many channels.

The fade curve follows a dB scale weighted towards the useful range: about three times as many steps are spent between −30 dB and 0 dB as below −30 dB. A fade from −∞ therefore passes the inaudible part quickly and moves smoothly through the range you actually hear.

Fade + Mute coordination

If a cue fades a fader and changes its mute state at the same time, the mute is timed intelligently:

- **Fading down into a mute:** the mute is applied right at the end of the fade — you hear the level drop away before the channel goes silent.
- **Fading up out of a mute:** the channel is unmuted immediately, so the fade-in is audible from the start.

GO during a fade, and Cancel fade

- Pressing **GO** on a new cue stops fades still running from the previous cue where they are; the new cue's fades start from those current positions.
- Pressing **Cancel fade** stops all running fades and jumps every fading parameter straight to its target value.
- The Cancel fade button counts down the seconds left on the newest fade.

Auto Update ignores the console's echo of a running fade, so intermediate fade steps are never written into your cues — only the fade's target value counts.

8. Auto Update

Auto Update keeps your cues in sync with live changes you make on the console — so if you rode a fader during a scene, the next time that cue is recalled it remembers where you left it, rather than snapping back to the originally saved value.

- Toggle **AUTO UPDATE** on/off in the connection bar, or remotely via OSC/TCP
- Configure it in the **Sections & Auto-Update** tab

Sections are required

Auto Update writes nothing until at least one Section exists in the show (the Sections list says "No sections → AU writes nothing"). Click **Add Section** to create one.

The **section selected in the Sections list** is the active one — its settings decide what happens to each change. Select a different section to switch behaviour.

What a Section controls

For each parameter type (one row per Recall Scope column, plus FX Racks), choose one of four columns:

Column	What happens to a live change
No Snapshots	Tracked live, but never written to any cue — whatever is programmed stays exactly as it is
Current Snapshot	Only the active cue is updated (like a DiGiCo exclusion)
Current Group	Every cue in the same group as the active cue is updated. If the active cue has no group, only the active cue is updated
All Snapshots	Every cue in the show is updated

New sections start with Fader, Mute, Sends and FX Racks set to Current Snapshot and everything else set to All Snapshots — level rides stay local to the current cue, while setup changes propagate everywhere.

Under **Channel Assignment** you can limit a section to specific inputs, buses, matrices, mains, DCAs and FX racks (Select All / Select None per tab). Changes on channels outside the section are not written.

Only parameters in the cue's Recall Scope are written

A change is only written to a cue if that parameter is included in the cue's Recall Scope. Excluded parameters are never touched.

A cue must be active

Current Snapshot and Current Group write to the **active** cue — the last cue fired with GO (or the cue just created with Add Snap). Selecting a cue in the list or using Recall to Wing does not change the active cue. Before any cue is active, nothing is written.

The status bar tells you what happened

With Auto Update on, every change you make on the console is reported in the status bar with a readable name, e.g.:

- AU: Input Ch 05 · Fader = −4.5 — saved to cue 003 'Playground'
- AU: Bus 03 · Sends (send MX1 lvl) = −10 — saved to 12 cues (all snapshots)
- AU: Input Ch 05 · EQ — not saved: set to 'No Snapshots' in section 'Band'
- … not saved: no active cue (press GO on a cue first)
- … not saved: channel isn't in the active section
- … not saved: excluded from Recall Scope in the target cue(s)
- … ignored, cue fade in progress

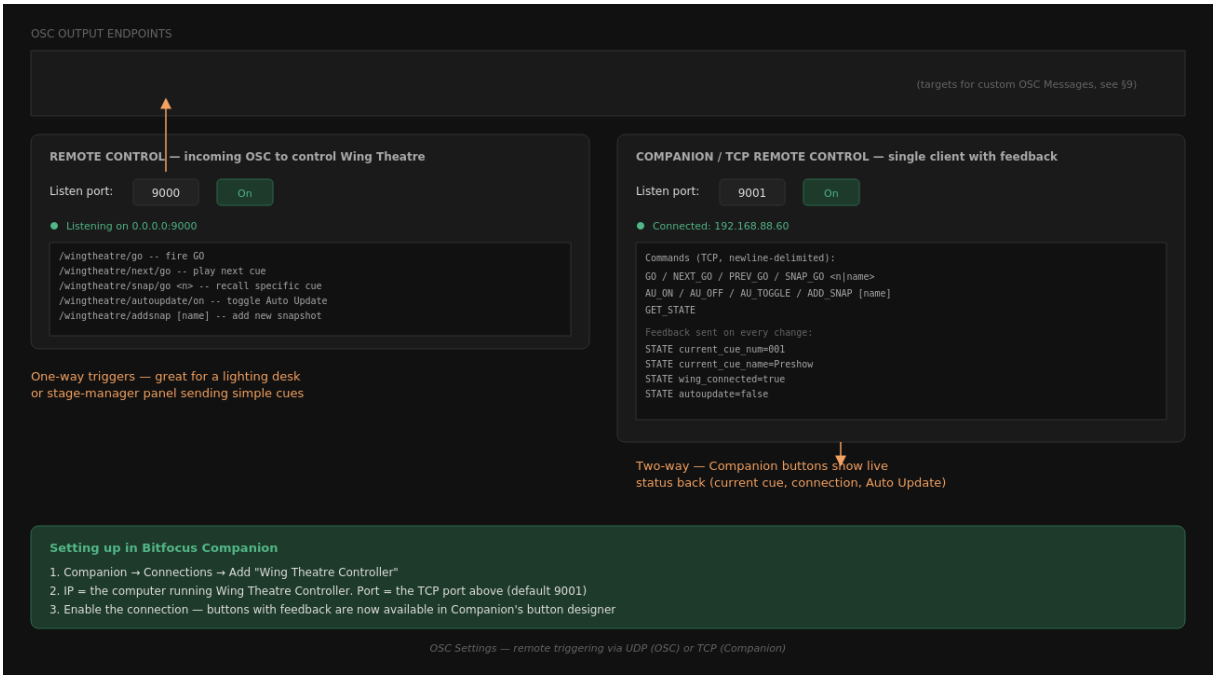
If something isn't being saved, the status bar always says why. All messages are also written to wing_theatre_log.txt.

9. Groups

Cues can be assigned to a named Group — in the Snapshot Details panel (pick a group or + Add group) or for many cues at once via right-click → Set group tag… in the cue list.

Groups are useful for organising a large show (e.g. all cues belonging to one song or scene), and they drive Auto Update's **Current Group** mode: a change made while a cue in the group is active is written to every cue in that group.

10. OSC Settings



OSC and TCP remote control settings

OSC Output Endpoints

External OSC targets (other consoles, media servers, lighting desks) that receive messages when a cue recalls, if that cue has custom messages defined in its OSC Messages scope tab.

- **+ Add Endpoint** — add a target (Name, IP Address, Port, Via interface)
- Toggle Enabled per endpoint
- **Remove Selected** to delete

Remote Control (UDP) — incoming OSC

Lets an external device (lighting console, stage manager panel, etc.) drive Wing Theatre Controller over OSC.

- Listen port — UDP port to listen on (default 9000)
- Toggle On/Off

OSC Address	Action
/wingtheatre/go	Fire GO (play current/selected cue)
/wingtheatre/previous/go	Play the cue before the last-fired one
/wingtheatre/next/go	Play the cue after the last-fired one
/wingtheatre/previous	Move selection up (no recall)
/wingtheatre/next	Move selection down (no recall)
/wingtheatre/snap/go <n|name>	Recall a specific cue by number or name

/wingtheatre/store	Update from Wing on the selected cue
/wingtheatre/autoupdate/on|off	Turn Auto Update on/off
/wingtheatre/addsnap [name]	Add a new snapshot (optional name)

Companion / TCP Remote Control — incoming TCP

A dedicated TCP protocol used by the Bitfocus Companion module, offering two-way state feedback (not just one-way triggering like OSC).

- Listen port — TCP port to listen on (default 9001)
- Toggle On/Off
- Up to 4 Companion connections at the same time (e.g. FOH, prompt desk and a backup). Every connection can send commands, and all of them receive the same feedback. A 5th connection is refused — an existing one is never disconnected to make room.

Command	Action
GO	Fire the currently selected cue
NEXT_GO	Fire the next cue
PREV_GO	Fire the previous cue
SNAP_GO <n|name>	Fire a specific cue
NEXT	Move selection down (no recall)
PREV	Move selection up (no recall)
AU_ON / AU_OFF / AU_TOGGLE	Control Auto Update
ADD_SNAP [name]	Add a snapshot
GET_STATE	Request a full state refresh

Feedback sent automatically on every change: current_cue_num, current_cue_name, next_cue_num, next_cue_name, selected_cue_num, selected_cue_name, autoupdate, wing_connected, cue_count, fading

Remote GO, Add Snap and Store follow the same rules as the buttons: they are blocked (with a status-bar message) until the Wing is connected and synced.

11. Bitfocus Companion Integration

A dedicated Companion module (v5+) ships alongside the app, giving you buttons with live feedback — the button text/colour updates automatically to reflect the current cue, connection state, and Auto Update status.

Installing the module

Companion loads third-party modules from a folder on disk — the module isn't installed through Companion's own interface, so this is a one-time manual setup per machine running Companion:

- Download companion-module-wingtheatre.zip from the Releases page and extract it anywhere (e.g. a companion-modules folder)

- Install the module's dependencies using Companion's own bundled Node.js — it doesn't need a separate Node/Yarn install:
    - Windows: double-click install-windows.bat inside the extracted folder — it finds Companion automatically and installs everything
    - macOS: open Terminal in the extracted folder and run:

```
/Applications/Companion.app/Contents/Resources/node-runtimes/node22/bin/node \
  /Applications/Companion.app/Contents/Resources/node-runtimes/node22/lib/
node_modules/corepack/dist/corepack.js \
  yarn install
```

- Open Companion Launcher (not the browser UI) → the gear/cog icon → Advanced Settings
- Under Developer, click Select and choose the PARENT folder that contains the extracted module folder (not the module folder itself)
- Enable Developer Modules, then click Launch GUI

The module now appears as an available connection type in Companion, and stays available across restarts — this only needs doing once per machine, and again after installing a newer version of the module.

Adding the connection

- In Companion → Connections, add Wing Theatre Controller
- Set the IP address to the computer running Wing Theatre Controller, and the port to the TCP port set in OSC Settings (default 9001)
- Enable the connection

Available Actions

GO, Next GO, Previous GO, Snap GO, Auto Update ON/OFF/Toggle, Next cue (no GO), Previous cue (no GO), Add Snap, Get State

Available Variables

$(wingtheatre:current_cue_num), current_cue_name, next_cue_num, next_cue_name, selected_cue_num, selected_cue_name, autoupdate, wing_connected, cue_count, fading

Available Feedbacks

autoupdate, wing_connected, fading, current_cue_num, current_cue_name — use these to colour-code buttons (e.g. red when Wing is disconnected, green when Auto Update is on).

Note: Companion talks to Wing Theatre Controller, not directly to the Wing console — Wing Theatre Controller is always the single source of truth in the middle.

12. Show Files

- **New Show** — start a blank cue list
- **Open... / Save / Save As...** — standard show-file management (.wts files). You're asked to save unsaved changes before opening another show or quitting; the window title shows * when there are unsaved changes.
- **Autosave** — choose an interval (0–60 minutes) in the toolbar; 0 disables it

What a show file contains

The full cue list with every cue's snapshot data, Recall Scope (including individual sends), fade times, groups, notes and OSC messages; the sections; the OSC endpoints; and app settings: Wing IP, network interface, remote-control ports and on/off state, autosave interval, the default Recall Scope, and the window's size and position.

When you open a show, the window returns to the size and position it was saved with. If that position isn't on any connected screen (e.g. a show saved with an external monitor), the window is centred on the main screen instead.

Safe saving

- Every save is written to a temporary file first and only then swapped in for the real file. If saving fails (disk full, power cut, crash), the previous version of the show file is left untouched.
- Autosave needs a file to write to: a show that has never been saved is skipped, with the status-bar message "Autosave skipped — save the show once (Save As)...". Use Save As once at the start.
- Autosave never runs in the middle of a fade — it waits until the fades are finished, so a save can never make a fade stutter.

Tip: keep numbered copies of your show (Save As ...05, ...06, ...) at key points in the rehearsal process.

13. Log Files, About and Version

Log files

Two text files are written in the same folder as the app (on macOS: next to Wing Theatre Controller.app; on Windows: next to the exe):

- **wing_theatre_log.txt** — a timestamped copy of every status-bar message: connection and sync events, Auto Update results, fade information, errors. A new "Session started" line marks each launch.
- **crash_log.txt** — full technical details if something unexpected goes wrong. The app tries to keep running after logging the error.

When reporting a problem, send both files along with the version number.

About

Click the **?** button at the far left of the toolbar to see the installed version number and build date, plus credits and a link to the project's GitHub page. Always include this version string when reporting an issue.

14. Troubleshooting

Symptom	Likely cause / fix
Won't connect to Wing	Wrong IP, wrong network interface selected in via, or Wing unreachable on the network. On macOS, check Local Network permission. See wing_theatre_log.txt.
GO, Add Snap, Update from Wing are greyed out	Not connected, or the sync hasn't finished yet. Wait for "✓ Wing connected — N parameters synced". If it never comes, disconnect and connect again.
GO does nothing on the console	Check the cue's Recall Scope isn't fully excluded, and that the cue has data (status bar: "NO DATA — run 'Update from Wing' first").
A parameter recalls to the wrong value	Check it's included in that cue's Recall Scope — an excluded parameter is always left as-is. Check whether Auto Update has written a newer value (status bar / log).

Auto Update doesn't save a change	Read the status-bar message — it always says why: no section, no active cue (press GO), channel not in the active section, set to No Snapshots, or excluded from Recall Scope.
A fade time I typed didn't apply	Use a number such as 20, 20.0 or 20,5. An empty channel cell inherits the group row's time.
Fade is still running and I need it done now	Press Cancel fade — all fades jump to their target values.
Channel names missing in Recall Scope	Names come from the console after the sync. Offline, the names stored in the selected cue are shown.
Connection lost during the show	Restore the network, press Connect and wait for the sync to complete before pressing GO.
Companion shows no connection	Confirm the TCP port matches on both ends, that TCP remote is switched on, and that the firewall allowed the app through. If 4 Companions are already connected, a 5th is refused (status bar: connection refused).
macOS says the app can't be opened	See §2: right-click → Open, or System Settings → Privacy & Security → Open Anyway.
Windows zip won't open on a Mac	Expected — the Windows zip is for Windows. Extract it on the Windows PC.
Window opens off-screen or too big	Fixed automatically on open; otherwise move/resize it and save — the new size is stored in the show.

Good to know

- Avoid commas in channel and cue names on the console — a comma can stop that name from being recalled correctly.
- Up to 4 Companion (TCP) connections are served at the same time; the OSC Settings tab shows how many are connected and from where.
- Show files are compatible across macOS and Windows.

15. Credits

- Wing Theatre Controller — Mikkel Peter Larsen & Claude (Anthropic)
- wingmon (Wing protocol bridge) — Mikkel Peter Larsen & Claude, built on libwing by dannyfiresnake
- Protocol documentation — Patrick-Gilles Maillot
- python-osc — attwad
- PyQt6 — Riverbank Computing

Source and releases: github.com/mikk3722/wing-theatre-controller